

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Using data analytics to enhance feedback quality and timeliness
<i>Company:</i>	University of Nottingham

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☒
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	machine learning and data science

Project Background Information

Assessments, including coursework and projects, are widely submitted, marked and returned to students digitally in HE institutions. This data contains critical information on the quality and timeliness of marking, the marking process, and how feedback is used to help students understand how they performed and how they can improve.

However, despite the critical value of feedback in the learning process, institutions have little understanding of the quality and consistency of marking and feedback. Providing academics, education leaders and students with a clear understanding of the marking and feedback process is critical to ensure a continuous improvement of practice.

¹ An overview of different research areas in the school of computer science can be found [here](#).

Requirements

The group must provide:

- A process that enables live, automated extraction of assessment and feedback data from the VLE (submission dates, marking dates, feedback release dates, grades and feedback text), respecting GDPR and university data-governance policy given the personal data involved.
- A data pipeline that cleans, validates, pseudonymises and models this data into a structure suitable for analytics and reporting.
- A student-facing dashboard showing individual turnaround times for marking and feedback, compared against the published feedback policy (e.g. 15 working days) and against anonymised cohort/module averages.
- A staff-facing dashboard showing marking timeliness and grade/feedback patterns for their own module(s), benchmarked against faculty and university-wide norms.
- A leadership dashboard aggregating timeliness and feedback-quality indicators at faculty, school, module and individual staff level, with drill-down and trend-over-time views.
- Role-based access control ensuring students see only their own data, staff see only their own modules, and leaders see appropriately aggregated/anonymised views.
- Clear documentation of the data model, ETL process and dashboard definitions, and a testing strategy covering data accuracy and access control.

Advanced / Stretch Objectives

Beyond the core requirements above, the group is encouraged to explore one or more of the following to extend the project's ambition:

- Automated feedback-quality scoring: use NLP/LLM techniques to assess feedback text against rubric criteria (e.g. specificity, actionability, alignment with learning outcomes) rather than relying on timeliness alone.
- Predictive analytics/early-warning system: model workload and historical patterns to flag modules or staff at risk of breaching turnaround targets before the deadline is missed.
- Anomaly and outlier detection for academic leaders: automatically surface modules or staff whose marking timeliness or feedback quality diverges significantly from faculty/university norms.
- Real-time/event-driven architecture: move beyond batch extraction to a live, event-driven pipeline (e.g. via VLE webhooks/API streaming) so dashboards update as marking happens.
- Longitudinal and comparative analytics: track trends over multiple years/cohorts, and support what-if or benchmarking comparisons across faculties or peer institutions.
- Conversational/natural-language interface: allow students or staff to query their dashboard in plain English (e.g. 'How does my feedback turnaround compare to last semester?').
- Recommendations engine: suggest targeted staff-development resources or process changes based on identified patterns in marking timeliness or feedback quality.

Skills	
<i>Highly desirable:</i>	<i>Desirable:</i>
• Python or R for data manipulation and analysis • SQL for querying relational databases • Experience with a BI/dashboarding tool (Power BI, Tableau, or similar) • Understanding of data privacy/GDPR principles when handling personal data • Basic NLP or text-analytics experience (for feedback-quality analysis)	• ETL/data-pipeline tools (e.g. Airflow, dbt) • API integration experience (VLE/LMS APIs in Moodle) • UX/UI design sensibility for dashboard usability • Familiarity with LLM/prompt-based tooling for text scoring
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.